§ 6 Angular

角动量算符 $[J_i, J_j] = i\hbar \varepsilon_{ijk} J_k$

1. 本征值/态：

$$[J^2, J_k] = \sum_j [J_j^2, J_i] = \sum_j J_j [J_j, J_i] + [J_j, J_i] J_j$$

$$= i\hbar \sum_{jk} \varepsilon_{ijk} (J_j J_k + J_k J_j) = 0$$

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J^2, J_z 的共同本征态 $|\lambda, j_z\rangle$

$$J^2 |\lambda, j_z\rangle = \lambda |\lambda, j_z\rangle, \quad J_z |\lambda, j_z\rangle = j_z |\lambda, j_z\rangle$$

升降算符：

$$J_\pm = J_x \pm i J_y$$

$$J_z (J_\pm |\lambda, j_z\rangle) = J_\pm (J_z |\lambda, j_z\rangle) + (J_z J_\pm - J_\pm J_z) |\lambda, j_z\rangle$$

$$= J_\pm j_z |\lambda, j_z\rangle \pm \hbar J_\pm |\lambda, j_z\rangle$$

$$= (j_z \pm \hbar) J_\pm |\lambda, j_z\rangle$$

$$\to J_\pm |\lambda, j_z\rangle = C_\pm |\lambda, j_z\rangle$$

定义 $j_z = m\hbar$, $|\lambda, m\rangle$

$$J_\pm |\lambda, m\rangle = C_\pm(\lambda, m) |\lambda, m \pm 1\rangle$$

归一化：$\langle \lambda, m \pm 1 | \lambda, m \pm 1 \rangle |C_\pm(\lambda, m)|^2 = |C_\pm(\lambda, m)|^2$

$$= \langle \lambda, m | J_\pm^\dagger J_\pm | \lambda, m \rangle$$

$$= \langle \lambda, m | (J_x \mp i J_y)(J_x \pm i J_y) | \lambda, m \rangle$$

$$= \langle \lambda, m | J_x^2 + J_y^2 \pm i [J_x, J_y] | \lambda, m \rangle$$

$$= \langle \lambda, m | J^2 - J_z^2 \mp \hbar J_z | \lambda, m \rangle = \lambda^2 - \hbar^2 (m \pm 1) m$$

令 $\lambda = j(j+1)\hbar$

$$C_\pm(j, m) = \hbar \sqrt{j(j+1) - m(m \pm 1)}$$

$m = j, j-1, \cdots, -j$

2. 角动量表象下的角动量算符：

$\langle jm'|J_z|jm\rangle = m\hbar\,\delta_{mm'}$

$\langle jm'|J^2|jm\rangle = j(j+1)\hbar^2\,\delta_{mm'}$

$\langle jm'|J_\pm|jm\rangle = C_\pm(j,m)\,\delta_{m',m\pm1}$

分别可以解出 J_x, J_y, J_z

$\left[\frac{3}{4} - \frac{1}{2}\left(-\frac{1}{2}\right)\right] = 1$

$-\frac{1}{2} \quad -\frac{3}{2}+1$

① $j=\frac{1}{2}$

$J_z = \frac{1}{2}\hbar\begin{pmatrix}1 & \\ & -1\end{pmatrix}$, $J^2 = \frac{3}{4}\hbar^2\begin{pmatrix}1 & \\ & 1\end{pmatrix}$

$J_\pm = \hbar\begin{pmatrix} & C_-(\frac{1}{2},\frac{1}{2}) \\ C_+(\frac{1}{2},-\frac{1}{2}) & \end{pmatrix}$ / $\hbar\begin{pmatrix} & C_-(\frac{1}{2},\frac{1}{2}) \\ C_+(\frac{1}{2},-\frac{1}{2}) & \end{pmatrix}$

$= \hbar\begin{pmatrix}0 & 0\\ 1 & 0\end{pmatrix}$ / $\hbar\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix}$

$\Rightarrow J_x = \frac{1}{2}(J_+ + J_-) = \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$, $J_y = \frac{1}{2i}(J_+ - J_-) = \frac{\hbar}{2}\begin{pmatrix}0 & i\\ -i & 0\end{pmatrix}$

② $j=1$

$J_z = \hbar\begin{pmatrix}1 & & \\ & 0 & \\ & & -1\end{pmatrix}$, $J^2 = \hbar^2\begin{pmatrix}2 & & \\ & 2 & \\ & & 2\end{pmatrix}$

$J_\pm = \hbar\begin{pmatrix} & C_+(1,0) & \\ & & C_+(1,-1) \\ & & \end{pmatrix} = \hbar\begin{pmatrix} & \sqrt{2} & \\ & & \sqrt{2} \\ & & \end{pmatrix}$

/ $\hbar\begin{pmatrix} & C_-(1,1) & \\ & & C_-(1,0) \\ & & \end{pmatrix} = \hbar\begin{pmatrix} & \sqrt{2} & \\ & & \sqrt{2} \\ & & \end{pmatrix}$

$\Rightarrow J_x = \frac{\sqrt{2}}{2}\hbar\begin{pmatrix}0 & 1 & \\ 1 & & 1 \\ & 1 & \end{pmatrix}$, $J_y = \frac{\sqrt{2}}{2}\hbar\begin{pmatrix}0 & -i & \\ i & & -i \\ & i & \end{pmatrix}$